

JEE MAIN REVISION MASTER: REDOX REACTIONS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 71 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Redox Reactions — Applied Cases, Reactions & Traps	49 PYQs	Applied Topics
Very High	Redox Reactions — Standard Calculations & Problem Archetypes	27 PYQs	Quantitative Analysis
High	Redox Reactions — Core Principles & Governing Laws	9 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Redox Reactions — Applied Cases, Reactions & Traps	$\text{H}_2\text{O}_2, \text{ClO}_3^-, \text{P}_4, \text{Cl}_2, \text{Cu}^{+1}, \text{NO}_2, \quad \text{K}_2\text{Cr}_2\text{O}_7, \quad 2\text{IO}_3^- + \text{I}^- + 12\text{H}^+ \rightarrow 6\text{I}_2 + 6\text{H}_2\text{O}$
Redox Reactions — Standard Calculations & Problem Archetypes	$\frac{10}{5}, \quad 2\text{IO}_3^- + \text{I}^- + 12\text{H}^+ \rightarrow 6\text{I}_2 + 6\text{H}_2\text{O}, \quad \text{KMnO}_4$
Redox Reactions — Core Principles & Governing Laws	$=20\text{ml}, \quad \text{Ca(OH)}_2, \quad \text{Sn}^{4+}, \text{Sn}^{2+}, \text{Pb}^{2+}, \text{Ti}^{3+}, \text{Pb}^{4+}, \text{Ti}^{+}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Redox Reactions — Applied Cases, Reactions & Traps (Very High Yield — 49 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Redox Reactions.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "In order to oxidise a mixture of 1 mole each of $\mathrm{FeC}_2\mathrm{O}_4$, $\mathrm{Fe}_2\mathrm{C}_2\mathrm{O}_4$, $\mathrm{Fe}_2\mathrm{C}_2\mathrm{O}_4$, $\mathrm{Fe}_2\mathrm{C}_2\mathrm{O}_4$..." [Ans: B]
- [2026 Evening Shift] Q#1 (Numerical): "200 cc of $x \times 10^{-3} \text{M}$ potassium dichromate is required to oxidise 750 cc of 0.6 M Mohr's salt solution in acidic medium. Here $x = \dots$ " [Ans: 375]
- [2026 Morning Shift] Q#2 (MCQ): "Given below are two statements: Statement (I): Oxidising power of halogens decreases in the order $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$, which is the ba..." [Ans: N/A]

3.2 Redox Reactions — Standard Calculations & Problem Archetypes (Very High Yield — 27 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Redox Reactions.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Evening Shift] Q#3 (MCQ): "One mole of Cl_2 (g) was passed into 2 L of cold 2 M KOH solution. After the reaction, the concentrations of Cl^- ..." [Ans: N/A]
- [2025 Evening Shift] Q#1 (MCQ): "0.1 M solution of KI reacts with excess of H_2SO_4 and KIO_3 solutions. According to equation $5\text{I}^- + \text{IO}_3^- + 6\text{H}^+ \rightarrow \dots$ " [Ans: N/A]
- [2025 Morning Shift] Q#2 (MCQ): "Given below are two statements: Statement I: In the oxalic acid vs KMnO_4 (in the presence of dil H_2SO_4) titration the solution..." [Ans: N/A]

3.3 Redox Reactions — Core Principles & Governing Laws (High Yield — 9 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Redox Reactions.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Redox Reactions.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Redox Reactions PYQs.

Representative Source PYQs:

- [2026 Morning Shift] Q#2 (MCQ): "Given below are two statements: Statement (I): Oxidising power of halogens decreases in the order $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$, which is the ba..." [Ans: N/A]
- [2025 Morning Shift] Q#3 (Numerical): "Some CO_2 gas was kept in a sealed container at a pressure of 1 atm and at 273 K. This entire amount of CO_2 gas was later passed t..." [Ans: 45]
- [2024 Evening Shift] Q#3 (MCQ): "The number of ions from the following that are expected to behave as oxidising agent is : $\text{Sn}^{4+}, \text{Sn}^{2+}, \text{Pb}^{2+}, \text{Ti}^{3+}, \text{Pb}^{4+}, \text{Ti}^{+}$..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Morning Shift (Q#1)

In order to oxidise a mixture of 1 mole each of FeC_2O_4 , $\text{Fe}_2(\text{C}_2\text{O}_4)_3$, FeSO_4 and $\text{Fe}_2(\text{SO}_4)_3$ in acidic medium, the number of moles of KMnO_4 required is

- (A) 3
- (B) 2 [Correct]
- (C) 5
- (D) 7

Correct Answer: B

Solution Analysis: In acidic medium, KMnO_4 acts as a strong oxidising agent. So, we need to find which species in the given mixture can be oxidised further. The mixture contains 1 mole each of: FeC_2O_4 , $\text{Fe}_2(\text{C}_2\text{O}_4)_3$, FeSO_4 , $\text{Fe}_2(\text{SO}_4)_3$. Now let us examine each compound. 1. FeC_2O_4 This is ferrous oxalate, so it contains: Fe^{2+} : can be oxidised to Fe^{3+} , $\text{C}_2\text{O}_4^{2-}$: can be oxidised to CO_2 . Oxidation of Fe^{2+} : $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + e^-$. So, 1 mole of Fe^{2+} requires 1 mole of KMnO_4 .

Benchmark Problem 2: JEE Main 2025 (Online) 4th April Morning Shift (Q#2)

KMnO_4 acts as an oxidising agent in acidic medium. 'X' is the difference between the oxidation states of Mn in reactant and product. 'Y' is the number of 'd' electrons present in the brown red precipitate formed at the end of the acetate ion test with neutral ferric chloride. The value of X+Y is _____.

Correct Answer: 10

Solution Analysis: X is difference in oxidation state. $7 - 2 = 5$. So $X = 5$. $6\text{CH}_3\text{COO}^- + \text{Fe}^{3+} + \text{H}_2\text{O} \rightarrow [\text{Fe}(\text{OH})_2(\text{CH}_3\text{COO})_6]^{+} + 2\text{H}^+$ [Brown red ppt]. $[\text{Fe}(\text{OH})_2(\text{CH}_3\text{COO})_6]^{+} + 4\text{H}_2\text{O} \rightarrow \text{ppt}[\text{Fe}(\text{OH})_2(\text{CH}_3\text{COO})] + \text{CH}_3\text{COOH} + \text{H}^+$. $\text{Fe}^{3+} \Rightarrow 3d^5 4s^0$ contains 5 d electrons. So $Y = 5$. $X + Y = 5 + 5 = 10$.

Benchmark Problem 3: JEE Main 2025 (Online) 22nd January Morning Shift (Q#3)

Some CO_2 gas was kept in a sealed container at a pressure of 1 atm and at 273 K. This entire amount of CO_2 gas was later passed through an aqueous solution of $\text{Ca}(\text{OH})_2$. The excess unreacted $\text{Ca}(\text{OH})_2$ was later neutralized with 0.1 M of 40 mL HCl. If the volume of the sealed container of CO_2 was x, then x is _____ cm^3 (nearest integer). [Given : The entire amount of $\text{CO}_2(\text{g})$ reacted with exactly half the initial amount of $\text{Ca}(\text{OH})_2$ present in the aqueous solution.]

Correct Answer: 45

Solution Analysis: Let moles of $\text{CO}_2 = n$ moles of $\text{Ca}(\text{OH})_2$ total initially $= 2n$ excess $\text{Ca}(\text{OH})_2 = n$ gm equivalent of $\text{Ca}(\text{OH})_2 =$ gm equivalent of HCl $n \times 2 = 0.1 \times (40 / 1000) \times 1$ $n = 2 \times 10^{-3}$ Volume of $\text{CO}_2 = 2 \times 10^{-3} \times 22400 = 44.8 \text{ cm}^3$